Royal Devon University Healthcare NHS Foundation Trust

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Caudal Epidurals (Ultrasound Guided) within Orthopaedic Interface Service - Clinical Guidance

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SUMMARY

The purpose of this document is to outline the stages that need to be completed for the delivery of ultrasound guided injections within the Orthopaedic Interface service by a trained clinician for appropriate spinal conditions.

KEY POINTS

Pre-Referral Guidelines
Referral Guidelines
Pre-Procedure Guidelines
Procedure Guidelines
Post Procedure Guidelines

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1. INTRODUCTION

1.1 This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, and adaptation, to fit individual circumstances.

1.2 The delivery of ultrasound guided caudal epidurals via the Orthopaedic Interface Service (OIFS) has been developed to enhance patient care by reducing waiting time and reducing travel time for patients that are currently requiring therapeutic caudal epidural intervention for spinal and radicular leg pain.

1.3 The service is in collaboration with the Pain Team (Northern) and clinical time has been funded for the service to be overseen by a Consultant Pain Physician.

1.4 The procedure will be undertaken by a Physiotherapy Advanced Practitioner/Physiotherapy Consultant who has prior training ultrasound (CASE accredited MSK Sonography PgCert). Non-Medical Prescribing and undertaken a Corticosteroid Injection Therapy module (M-Level). Clinicians will have undertaken an appropriate period of training overseen by a Consultant Pain Physician where they deemed competent to perform all aspects of the procedure independently.

1.5 The procedure will only be undertaken when a Consultant Pain Physician is available to support and has had a briefing prior to the clinic starting.

2. BACKGROUND

2.1 The prevalence of spinal pain within primary care and musculoskeletal is high. Low back pain has increased to become the leading cause of disability-affected life years across all ages

2.2 There has been an increased demand for musculoskeletal (MSK) and spinal services nationally. This has also caused an increased demand on other interventional services such as radiology and pain clinics which provide caudal epidurals for patients with radicular pain.

2.3 Whilst many patients can be managed conservatively with exercise therapy, those with an acute presentation, with a high level of radicular pain, find it difficult to engage with physiotherapy and are often referred on to other pathways which support pain relief. Caudal epidurals are commonly used to manage acute spinal pain.

2.4 Effectiveness studies suggest that patients with acute pain have improved outcomes following caudal epidural with 70%–80% of patients reported to achieve more than 50% improvement in pain and function following injection.

2.5 The current guidance indicates that patients with acute (less than 3 months) or severe lumbar radicular pain can be offered caudal epidural interventions to help relieve their pain and increase mobility.

2.6 Inclusion criteria

- A current history of sciatica type pain with or without back pain with a clinical hypothesis that the pain is of spinal radicular origin.
- A lumbar-sacral spine MRI showing nerve root compression of discal or foraminal stenotic origin, which correlates with the patient's presenting symptoms

2.7 Exclusion criteria

- Unable to provide consent
- Recent or unresolved infection
- Central canal lumbar stenosis
- Allergic to medication or products used within the delivery of the caudal epidural
- Medication with a known adverse interaction with corticosteroids/local anaesthetic
- Previously suffered from arachnoiditis
- Patients taking anticoagulant, or anti-platelet medication that have not had documented prior approval for the procedure from their General Practitioner and or Consultant
- Uncontrolled diabetes or other uncontrolled medical conditions such as hypertension.
- Surgical implants in the sacrum or pelvis.
    - Patients with abnormal anatomy as found on MRI scan such as Tarlov cysts which may impede needle placement.

3. DEFINITIONS

3.1 ULTRASOUND-GUIDED CAUDAL EPIDURAL is a minimally invasive procedure used to deliver medications, such as local anaesthetics and steroids, into the epidural space at the base of the spine. The caudal approach accesses the epidural space through the sacral hiatus, located at the lower end of the sacrum.

3.2 STERILE FIELD is a designated area that is maintained free from microorganisms to prevent infection during medical or surgical procedures. It typically includes sterilised instruments, drapes, and supplies, as well as the prepped area of the patient's body. Strict aseptic techniques are used to establish and preserve the sterility of this field, ensuring a controlled environment for safe interventions.

3.3 EQ-5D-5L is a tool for measuring health-related quality of life. It evaluates five dimensions (mobility, self-care, usual activities, pain/discomfort, and anxiety/depression) across five severity levels. It also includes a visual analog scale (0–100) for overall health, with 0 being the worst and 100 the best imaginable health

4. MAIN BODY OF GUIDELINE

4.1 Pre-Referral Guidelines

- Prior to referral all patients will be clinically assessed by an OIFS Spinal Specialist and all essential imaging will have been undertaken.
- The patient’s case will have been discussed at the Orthopaedic Spinal/OIFS MDT and the appropriate outcome will be documented on EPIC.
- If the patient is on anticoagulation medication the referring clinician will discuss with the appropriate General Practitioner and/or Consultant if the patient is suitable for the procedure and document the appropriate starting and stopping regime of medication if required. This will be documented on EPIC.
- Patient information leaflet provided prior to the referral.

4.2 Referral Guidelines

Patients are seen in an outpatient appointment with the Spinal Interface team and referred to Caudal Epidural Injection clinic (CAUDEP).

- The referrer dictates a referral letter with details of the request, including:
  - Diagnosis
  - What is being requested and why
  - Any contradictions
  - 'CAUDAL EPIDURAL REFERRAL' to be written at the top of the letter
  -
- The letter is to be addressed to Ndh Orthopaedic Interface Medical Secretary, via in basket message on EPIC, as the primary recipient.

- The referral letter is then routed to the Orthopaedic Interface Service (OIFS) admin team (P Ndh Orthopaedic Interface Medical Secretary) via the letter queue on EPIC, for grammar and format review.

- The admin team will route the final version to appropriate in basket for triage.

- At the point of referral, the referring clinician will place an order for the patient for a 'NEW FACE-TO-FACE' appointment under the clinic code:
  34396 NDDH ORTHOPAEDIC INTERFACE SERVICE – CAUDEP – CLINIC

- Each referral will be triaged appropriately and either accepted or rejected.

- The triaging clinician will contact admin to inform them of the triage outcome, via an inbasket message, on EPIC. If the patient is accepted, a comment will be added to the waitlist saying 'ACCEPTED'. If the patient is rejected, the comments will be sent back to the referring clinician with the waitlist removed.

- The patient will be followed up 6 weeks post procedure by the referring clinician and it is their responsibility to place an order for a follow up.

- Any questions or queries regarding the patient and the CAUDEP appointment should be directed to the original referrer.

- At the point of referral the patient will be asked to complete an EQ5D5L spinal outcome measure.

4.3 Pre-Procedure Guidelines

- Team briefing will be undertaken to ensure all patients on the day are discussed

- Consultant Pain Physician will be informed of clinic list

- On arrival to the procedural room patient's full name and date of birth will be confirmed

- Blood pressure with be monitored to detect unexpected hypertension (140/90). Pulse and temperature will be monitored to detect any potential infection or uncontrolled tachycardia.

- The risks of the procedure will be fully discussed and written consent obtained

- Prior to the procedure World Health Organisation pre procedure checks will be carried out.

4.4 Procedure Guidelines

- Patient placed in prone position and lower back/sacrum exposed to ensure injection site is clearly visible. The injection site is cleaned twice with Chlorhexidine 2%.

- The procedure will be done under sterile conditions.

- Medication is drawn up under sterile conditions
  - Skin block
    - 1% lidocaine
    - 25g orange needle
    - 5ml syringe
  - Spinal injection – only NRfit equipment to be used
    - Depo-Medrone 40 mg 1ml
    - 0.5% Levobupivacaine or equivalent 3ml
    - 0.9% Sodium Chloride 6ml
    - 10ml syringe
    - Pujunk Sonoplex needle 22 gauge 80 mm
  - Appropriate filter needles will be used for glass ampoules

- The cleaned ultrasound probe is covered with appropriate probe cover.

- A sterile drape will be used around exposed injection site.

- Sterile gel will be applied to skin, and the sacral hiatus will be visualised in both long and short axis.

- Prior to preforming procedure Stop Check Block will be undertaken to confirm procedure and patient's full name.

- Skin block with local anaesthetic to a maximum of 5mls 1% lidocaine prior to caudal epidural with Pajunk 80mm NrFit needle under direct US guidance.

- Any bleeding controlled and Opsite spray applied to the injection site.

- The patient will be given the opportunity to slowly return to sitting position when they are ready

4.5 Post Procedure Guidelines

- Post procedure World Health Organisation post procedure checks will be carried out.
- The patient will be monitored in the clinic room to ensure they are fit and well to be able to leave.
- A GP discharge summary will be generated via EPIC and sent within 24 hours
- Aftercare will be discussed with the patient. Patients will be able to contact the Orthopaedic Interface Service within office hours via 01271 311565 if there are any post procedure concerns.
- The referring clinician will be informed that the procedure has been undertaken, and a six week follow up order will be placed by the referring clinician.
- At the six week follow up the patient will be asked to complete a post procedure EQ5D5L spinal outcome measure.
- If any unexpected symptoms arise post procedure these will be discussed immediately with the Consultant Pain Physician.
- In the event of a critical event post procedure the 2222 call will be made in keeping with current Therapies Department practice and the Consultant Pain Physician informed immediately
- A verbal team brief will be undertaken by the senior clinician with all team members present and will include:
    - Things that went well
    - Any problems with equipment and other issues
    - Areas for improvement
    -
- If required any issues will be escalated to the appropriate member of staff

5. MONITORING COMPLIANCE WITH THIS GUIDELINE

5.1 Monitoring

- Clinical Audits: Regular reviews of referrals, procedure adherence, and documentation.
- Procedure Reviews: Random quarterly checks of 10% of procedures for compliance.
- Patient Feedback: Surveys to gather satisfaction and outcome data.
- Team Briefings: Routine pre- and post-clinic debriefs to identify issues and improvements.
- Incident Reporting: Mandatory reporting of adverse events or deviations for review.
- Training Records: Ensuring clinicians have completed required training and competency assessments.

5.2 Compliance

- Oversight: Consultant Pain Physician oversees adherence with support from OIFS admin.
- Documentation: All patient and procedural records must meet set standards.
- Escalation Protocols: Issues are escalated to Clinical Governance Leads.
- Continuous Improvement: Findings from audits, incidents, and feedback inform updates.
- Annual Review: Guidelines reviewed annually to reflect new evidence and learning.

6. ASSOCIATED CLINICAL GUIDELINES, REFERENCES, POLICIES/PROCEDURES

- Prevention of Inoculation Injuries
- Prepare and Maintain a Sterile Trolley and Circulating for Interventional Radiology Procedures (Competency Assessment).doc
- Acute anaphylaxis management SOP.pdf